Jezero Crater, Mars
NASA Research Data Pack

The single source the developer needs to design and build Jezero Crater: every relevant NASA orbital and surface image embedded with its metadata, elevations and topography, six zone build sheets with sample data tables, the verified chemicals, and a map build specification. Prepared by the team researcher. Expanded edition - nothing in the first edition has been removed.

> **NASA-first rule.** NASA data is the core of this pack and every design-critical number comes from it. Four clearly-labelled supplements the team agreed to allow - USGS mosaics built from NASA MRO data, DLR/HRSC topography, and mission-team journal results via NASA NTRS and AGU - are marked as supplements wherever they appear.

How the developer should use this pack

- **Section 1** - seven embedded NASA images (orbital, traverse maps, surface view, instrument data), each with its PIA number, spacecraft, instrument, date and credit, plus elevations and topography (Section 1.6) and the archive links for full-resolution build data (Section 1.5).
- **Section 2** - the scientific area and the map of the work area: a schematic plan, a zone master table, and a build sheet for each of the six zones (Section 2.4).
- **Section 3** - the chemicals and minerals by zone and elevation, with real sample records to copy for labels and data displays (Section 3.4).
- **Section 5** - the map build specification: coordinate frame, scale, layer stack, legend and the full label list. Work only from this pack; the links are for judges and for downloading originals, not for finding design data.
- Every fact carries its reference number, [R1] to [R31]. The full reference list is in Section 4.

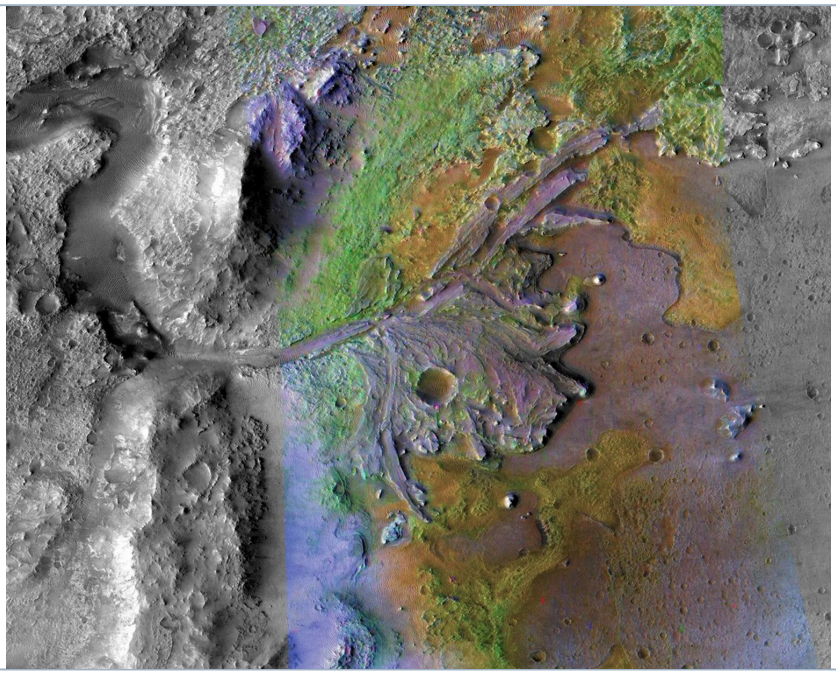
Jezero at a glance - all values as NASA publishes them

Location	Western edge of Isidis Planitia, a giant impact basin just north of the Martian equator, at about 18 degrees latitude, 77 degrees longitude [R2]. The THEMIS image in Section 1 is centred at latitude 18.449, longitude 77.436 [R4].
Diameter	28 miles (45 kilometres) wide [R1][R2]. Note: the NASA caption for PIA26140 states 47 km (29 miles) [R4]. NASA's own pages differ - see the cautions in Section 3. Do not silently pick one value.
Ancient lake	More than 3.5 billion years ago, river channels spilled over the crater wall and created a lake. Water carried clay minerals from the surrounding area into the crater lake [R1].
Rover	NASA's Perseverance rover, Mars 2020 mission. NASA examined more than 60 candidate landing locations before choosing Jezero. Perseverance landed in Jezero Crater on Feb. 18, 2021 [R1].
Mission goal	Searching for signs of ancient microbial life. The four science objectives are: studying Mars' habitability, seeking signs of past microbial life, collecting and caching samples, and preparing for future human missions [R1].

1. Orbital image data

Three NASA orbital products cover the three jobs the design needs: a whole-area base map that also carries mineral information, a high-detail view of the delta front, and a false-colour view of the western half that separates surface materials. The images printed here are the NASA Photojournal versions. For building, download the full-resolution products from the NASA links given with each image and from the archives in Section 1.5. Sections 1.4 and 1.6 add the traverse working maps and the elevation data the map is built from.

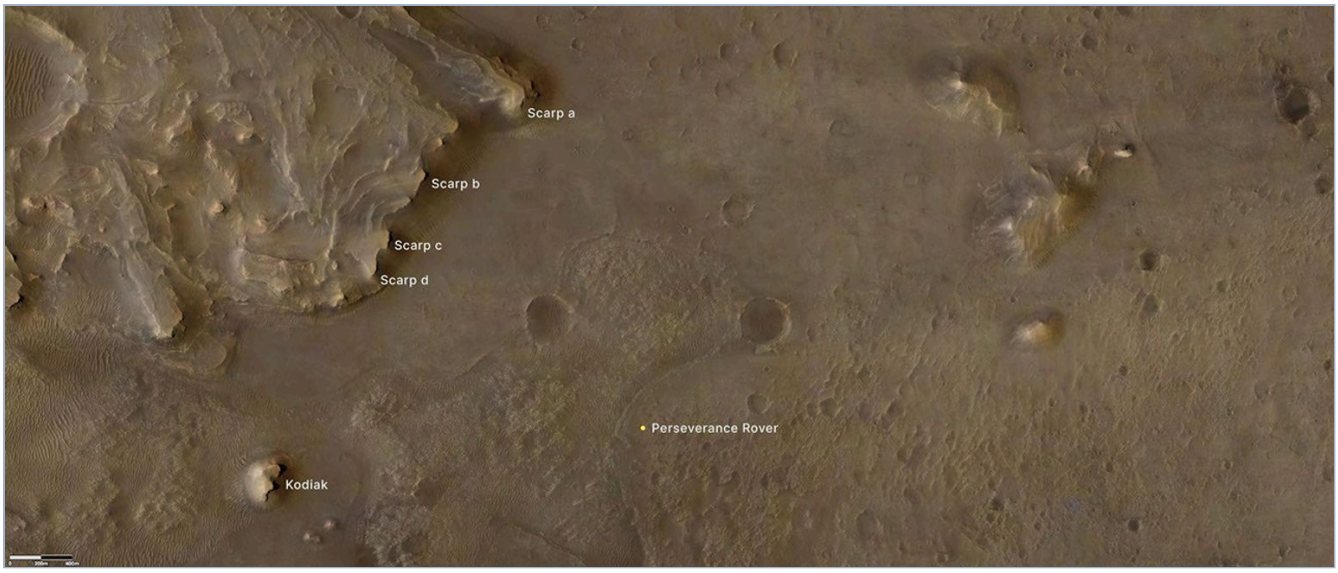
1.1 Base map - PIA23239: Jezero Crater, Mars 2020's landing site



PIA23239. Credit: NASA/JPL-Caltech/MSSS/JHU-APL. The colour overlay is CRISM mineral data; the greyscale base is CTX imagery. NASA's caption states that in the Jezero delta, sediments contain clays and carbonates [R3].

NASA identifier	PIA23239
Spacecraft	NASA Mars Reconnaissance Orbiter (MRO)
Instruments	Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) combined with the Context Camera (CTX). CRISM was built and is operated by the Johns Hopkins University Applied Physics Laboratory; CTX was built and is operated by Malin Space Science Systems [R3].
Published by NASA	2019-05-15
What it shows	Jezero Crater, the Mars 2020 landing site. Spectral data from orbit show minerals altered by water in the fan and delta sediments [R3].
Use it for	The master base map: crater outline, western delta fan, inlet channel, and a first mineral zoning from the CRISM colour overlay (clays and carbonates in the delta).
NASA reference	https://science.nasa.gov/photojournal/jezero-crater-mars-2020s-landing-site/ and https://photojournal.jpl.nasa.gov/catalog/PIA23239 [R3]

1.2 Delta detail - PIA24814: Jezero Crater's Kodiak and scarps

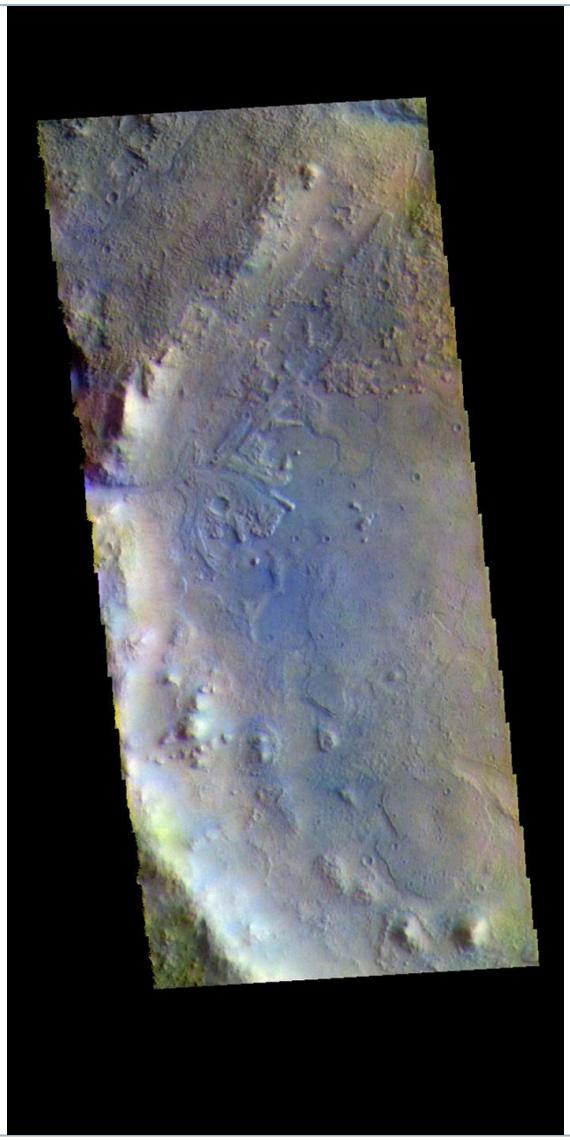


PIA24814. Credit: NASA/JPL-Caltech/University of Arizona/USGS. Labels are part of NASA's published image: Scarp a to Scarp d along the delta front, Kodiak at lower left, and the Perseverance rover position marked at lower right [R5].

NASA identifier	PIA24814
Spacecraft / instrument	NASA Mars Reconnaissance Orbiter / High Resolution Imaging Experiment (HiRISE), operated by the University of Arizona [R5]
Published by NASA	2021-10-07
What it shows	An overhead view of a portion of Jezero Crater. The yellow dot at lower right is the location of Perseverance. Kodiak, the remnant of the crater's delta, is at lower left. The long, steep slopes called scarps along the delta are at upper left, labelled A through D [R5].
Distances and detail	NASA's companion report adds that the scarps were about 1.2 miles (2.2 kilometres) northwest of the rover when imaged, with Kodiak at about the same distance to the southwest, and that the scarps contain boulders up to 5 feet (1.5 metres) across in their upper layers [R6].
Use it for	Sculpting the western delta front: the four scarps as steep layered banks, Kodiak as a separate flat-topped delta remnant, and the correct rover-to-delta spacing.
NASA reference	https://science.nasa.gov/photojournal/jezero-craters-kodiak-and-scarps/ [R5]

Why this image matters for the build. Before Perseverance arrived, Kodiak had been imaged only from orbit. The rover's surface images later revealed the layering along Kodiak's eastern face - the order and position of rock layers a geologist would expect in a river delta on Earth. NASA's lead author for that study called this the key observation confirming a lake and river delta at Jezero [R6]. Model Kodiak and the scarps as layered delta deposits, not as plain hills.

1.3 False colour, western half - PIA26140: Jezero Crater false colour



PIA26140. Credit: NASA/JPL-Caltech/ASU. False colour - read the caution on this page before using its colours [R4].

NASA identifier	PIA26140
Spacecraft / instrument	NASA 2001 Mars Odyssey / Thermal Emission Imaging System (THEMIS) VIS camera, 5 filters [R4]
Orbit number	95046
Centre latitude	18.449
Centre longitude	77.436
Captured	2023-05-19 09:28
Published by NASA	2023-09-29
What it shows	The western half of Jezero Crater. Perseverance is in this part of the crater, near the delta deposit formed by silt-laden water flowing into the crater during a wetter period in Mars' past. The crater most likely hosted a lake for a period of time [R4].
Caption diameter	This NASA caption states Jezero Crater is 47 km (29 miles) in diameter - see the diameter caution in Section 3 [R4].

False colour is not true colour. NASA states that the THEMIS filters do not span the full range of colours seen by the human eye, each filter image is contrast-enhanced before combination, and the apparent colour variation is therefore exaggerated - but it does represent a real change in the surface. The long edges of THEMIS colour images typically contain colour artefacts that do not represent surface variation [R4]. Use PIA26140 to zone materials, never as the literal ground colour.

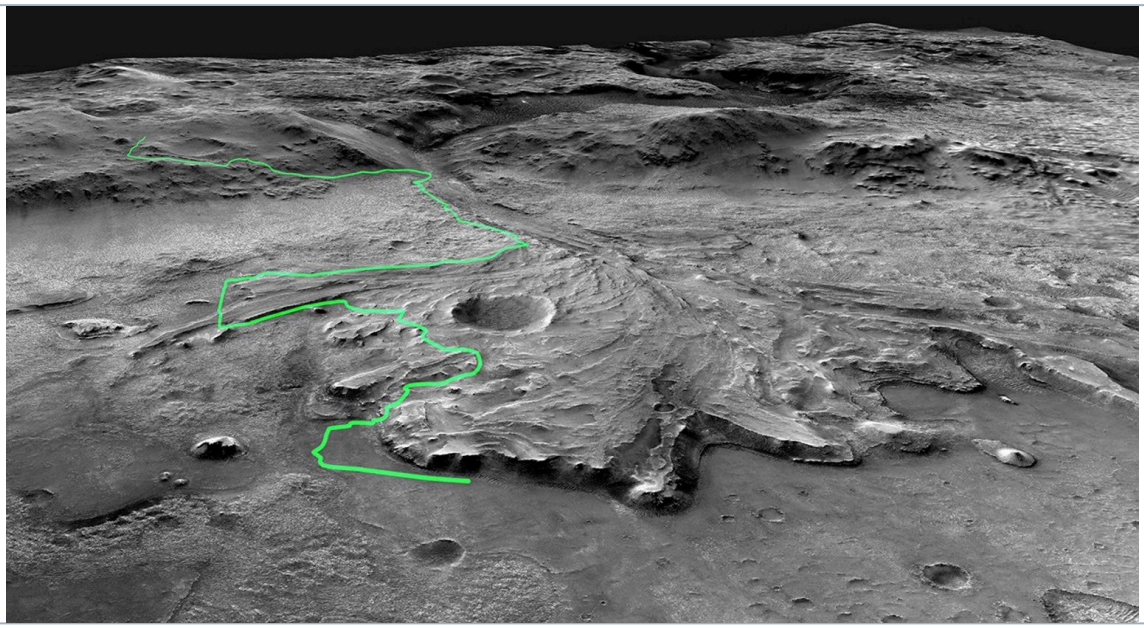
NASA reference: https://science.nasa.gov/photojournal/jezero-crater-false-color/ [R4]

How the colour image is made, per NASA: three of the five grayscale filter images are selected, each is contrast-enhanced and converted to a red, green or blue intensity image, and the three are combined into a single colour image [R4].

1.4 Traverse and route maps - the map-designer views

Sections 1.1 to 1.3 are the science images. The two products in this section are the working maps: they show how the zones connect, where the rover actually drove, and the true scale of the delta. Every later section refers back to these two figures.

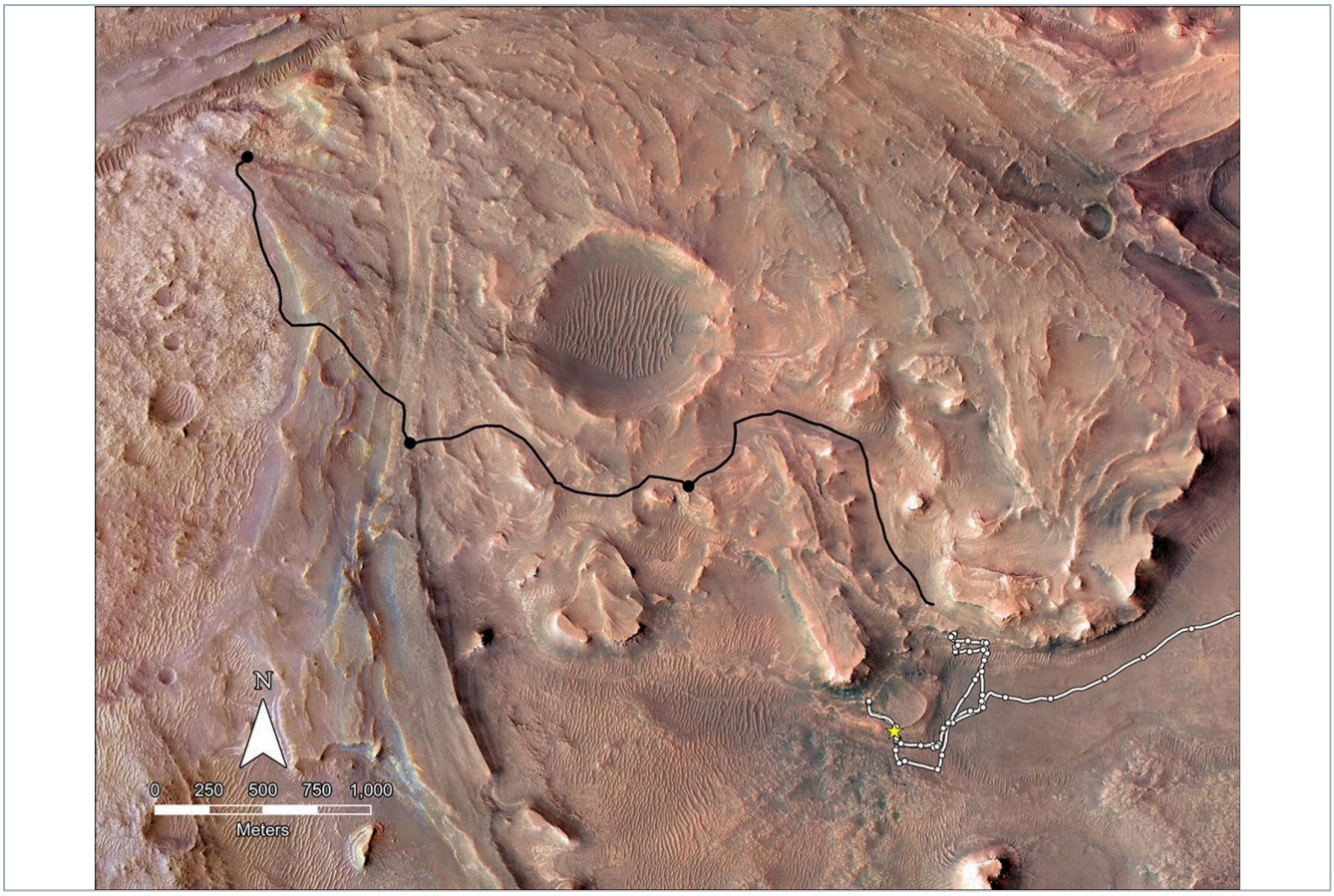
1.4a Whole-route perspective - PIA24379: Possible path for Perseverance rover



PIA24379. Credit: NASA/JPL-Caltech. Oblique perspective, not a flat map: the green line is a possible route drawn before landing, from the cliffs at the base of the delta, up and across the delta toward possible ancient shoreline deposits, then up the crater rim to the surrounding plains [R16].

NASA identifier	PIA24379
Spacecraft / instrument	NASA Mars Reconnaissance Orbiter / Context Camera (CTX). The mosaic is composed of multiple precisely aligned CTX images [R16].
Resolution	6 metres per pixel [R16]
Published by NASA	2021-01-27
Key scale facts	The route climbs the 2,000-foot-high (610-metre-high) crater rim. The prominent crater near the centre of the image is about 0.6 miles (1 kilometre) across - use it as an in-image scale check [R16].
Use it for	The 3D shape of the work area: delta front cliffs, fan surface, shoreline bench and rim wall in one view. This is the reference for how steep each zone should feel in a 3D build.
NASA reference	http://science.nasa.gov/photojournal/possible-path-for-perseverance-rover/ [R16]

1.4b Delta-top working map - PIA25673: Map of Perseverance's Delta Top Campaign



PIA25673. Credit: NASA/JPL-Caltech. White line and dots: ground already covered. Black line and dots: the planned 2023 route across the delta top, with notional science and sampling waypoints. The printed scale bar runs 0 to 1,000 metres and north is up [R17].

NASA identifier	PIA25673
Published by NASA	2022-12-15
What it shows	The top of Jezero's western delta at working-map scale: fan-top channels and layers, the crater embedded in the fan, and the fan front dropping to the crater floor at lower right [R17].
Use it for	Copy this map's conventions into the project map: north arrow top-left, a 1 km scale bar, white for completed traverse, black for planned route, dots for waypoints. Tile the delta-top texture from this view.
NASA reference	https://science.nasa.gov/photojournal/map-of-perseverances-delta-top-campaign/ [R17]

Live traverse. For the rover's latest position and full driven path, NASA maintains the Perseverance location map, built on a high-resolution colour base map from MRO HiRISE images, with each dot marking the end of a drive by sol [R29]. The printed maps in this pack are the fixed design references; the live map is the check for "where is the rover now."

1.5 Where to download the build data

The three Photojournal images are the design references. The sources below are the production data behind them: the NASA archives and mosaics the developer should pull geometry, elevation and rover imagery from. Record the product name and release in the project credits exactly as written here.

Source	What the developer gets	NASA link
NASA Planetary Data System (PDS) - Imaging Node portal	The archive entry point for MRO data. Its mission table lists Mars Reconnaissance Orbiter with instruments HiRISE, MARCI and CTX, with Atlas Product Search and Online Data Volumes. Search here for the individual Jezero observations behind the mosaics [R7].	https://pds-imaging.jpl.nasa.gov/portal/
PDS Geosciences Node - Mars 2020 mission page	All Perseverance science data, released three times a year by PDS. Bundles include PIXL, SHERLOC, SuperCam, RIMFAX, MEDA, MOXIE, engineering cameras and the Sample Dossier. Listed release 16 covers sols 1740-1859 (Jan. 10, 2026 - May 13, 2026) [R8].	https://pds-geosciences.wustl.edu/missions/mars2020/
Perseverance Raw Images	Unprocessed images sent back by the rover, browsable by sol and camera, including Mastcam-Z, Hazcam, Navcam and WATSON frames. Use these for ground-level texture and colour reference inside the crater [R1].	https://mars.nasa.gov/mars2020/multimedia/raw-images/index.cfm?begin_sol=907

Instrument key - what each NASA instrument contributes

Instrument	Spacecraft / rover	Contribution to this project
CTX - Context Camera	MRO	Wide orbital imagery and stereo terrain: the 6 m/pixel mosaic and DTM base [R9].
HiRISE	MRO	Highest-detail orbital views and stereo elevation: delta scarps, Kodiak, hazard-scale detail [R5][R10].
CRISM	MRO	Orbital mineral mapping from spectral data: clays and carbonates in the delta sediments [R3].
THEMIS VIS	Mars Odyssey	Five-filter false-colour imaging that separates surface materials in the western crater [R4].
SuperCam	Perseverance	Mast instrument that reads mineralogy from reflected light and fires a laser up to 21 feet (6.5 metres); analysed more than 185 bedrock targets in the Margin Unit [R11].
PIXL and SHERLOC	Perseverance	The two instruments that first collected the Cheyava Falls data: mineral reaction fronts and organic chemistry in Section 3 [R12].

1.5 (continued) Terrain mosaics and elevation models

Source	What the developer gets	Link
Mars 2020 Terrain Relative Navigation (TRN) CTX mosaic and DTM USGS-hosted, NASA data	A visible image mosaic of Jezero built from MRO Context Camera (CTX) images, orthorectified with digital terrain models (DTMs) from CTX stereo images, at approximately 6 metres per pixel - the resolution at which the CTX images were acquired. It was the onboard Lander Vision System map for landing and the ground-truth base for the HiRISE orthomosaic. Coverage: latitude 18.211 to 18.7212, longitude 77.1605 to 77.6992. Published 2020-07-24 [R9].	https://astrogeology.usgs.gov/search/map/Mars/Mars2020/JEZ_ctx_B_soc_008_orthoMosaic_6m_Eqc_latTs0_lon0
Mars Sample Return TRN HiRISE DTM and orthoimage mosaics USGS-hosted, NASA data	Digital elevation model and orthoimage mosaics of the Jezero crater floor built from 8 HiRISE stereo pairs, generated by NASA JPL in coordination with USGS Astrogeology in 2023-2024 for Mars Sample Return landing simulations. Horizontal alignment between the individual HiRISE DEMs is better than 0.9 m at the 99th percentile; vertical offset is under 2.3 m at the 99th percentile, except on the crater rim. Mosaic download 5.3 GB; individual images 13.2 GB. Published 2025-02-28, DOI 10.5066/P13CPYYU [R10].	https://astrogeology.usgs.gov/search/map/mars-sample-return-terrain-relative-navigation-hirise-dtm-mosaic

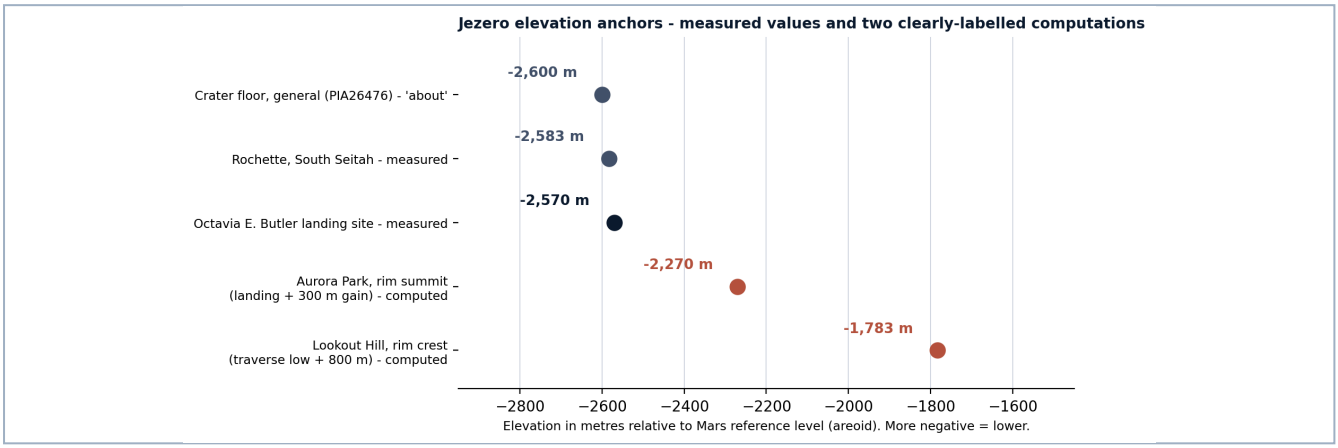
> **Build order that matches the data.** Start the terrain from the CTX DTM and 6 m/pixel CTX mosaic [R9]. Refine the delta, landing area and rim features the viewer will see up close with the HiRISE DTM mosaic [R10]. Texture the mineral zones from PIA23239 [R3]. Check ground-level appearance against Perseverance raw images before finalising colours.

Credit lines to copy into the project

- PIA23239 - NASA/JPL-Caltech/MSSS/JHU-APL
- PIA24814 - NASA/JPL-Caltech/University of Arizona/USGS
- PIA26140 - NASA/JPL-Caltech/ASU
- PIA24379, PIA25673, PIA26336, PIA26639 - NASA/JPL-Caltech
- CTX mosaic and DTM - NASA Mars Reconnaissance Orbiter CTX, via USGS Astrogeology Science Center
- HiRISE DTM mosaic - NASA JPL / USGS Astrogeology Science Center, DOI 10.5066/P13CPYYU

1.6 Elevations and topography

Jezero is a deep bowl. NASA's elevation graph for the mission, PIA26476, plots every SuperCam sampling site against elevation and states the datum plainly: Jezero Crater sits about 8,530 feet (2,600 metres) below Mars reference level - the areoid, the Martian equivalent of sea level - which is why the graph's numbers decrease as Perseverance gains elevation. Dotted lines on that graph mark estimated levels of the ancient lakes, and at its far right the line jumps sharply as the rover ascends toward the rim [R14].



Elevation anchors compiled for this pack. Dark dots are values reported from the rover's traverse; the general floor value is NASA's "about -2,600 m" [R14]. Rust dots are arithmetic from NASA's stated gains (+300 m, +800 m) and are labelled computed - they are not NASA-published elevations.

Sample elevation data - copy these values into the build

Point or span	Elevation / relief	Meaning for the build
Crater floor, general	About -2,600 m [R14]	Set the floor datum of the model here. All other heights are relative to this bowl.
Octavia E. Butler landing site	-2,570 m (reported from the first traverse)	The operations hub (Zone 6) sits 30 m above the general floor value - a gentle local rise, not a hill.
Rochette, South Seitah	-2,583 m (reported from the first traverse)	Low point of the floor leg: the whole crater-floor traverse varies by only about 13 m. Keep the floor almost flat.
Margin Unit explored span	265 m of elevation [R11]	The shoreline band is a long slope, not a cliff: spread its chemistry change over this full rise (Section 3.3).
Rim climb to Aurora Park	+300 m gain; slopes up to 23 degrees [R15]	The designed climb out of the crater. Rover drivers avoid tilts over 30 degrees - keep playable slopes at or below the 23-degree maximum NASA reports [R15].
Rim height above delta base	610 m [R16]	Full wall height to model on the western side, from delta-base cliffs to rim crest.
Lookout Hill crest	800 m above the traverse's lowest point [R22]	Total relief of the full route, floor to outside the rim. First views beyond the rim are to Nili Planum [R22].
Ancient lake depth (supplement)	At least 250 m deep; about 1,340 sq km [R27]	DLR/HRSC topography (non-NASA supplement): the waterline the Margin Unit and delta heights must agree with.

1.6 (continued) Terrain data products - resolution is the design budget

Product	Resolution	Build rule
CTX image mosaic (TRN) [R9]	About 6 m/pixel	Whole-crater texture. One pixel = a feature the size of a room; do not draw smaller detail from it.
CTX DTM mosaic [R24]	20 m/pixel posting; file name carries "20m" [R24]	Whole-crater terrain mesh. Vertical registration between the individual CTX DTMs is 3.8 m [R24].
HiRISE image mosaic [R25]	25 cm/pixel	Hero-area texture only (delta front, landing area, Bright Angel). Used by NASA to identify hazards in the landing ellipse [R25].
HiRISE DTMs	1 m posting (standard USGS HiRISE DTM grid)	Hero-area terrain. The Jezero HiRISE DTM mosaic's horizontal alignment is better than 0.9 m and vertical offset under 2.3 m at the 99th percentile, except on the rim [R10].
MOLA + HRSC context [R27]	Regional	Only for the country beyond the rim (Nili Planum, Isidis edge). Never for in-crater detail.

> **Why tiling is mandatory - computed.** At HiRISE's 25 cm/pixel, the 45 km crater would be 180,000 pixels across. No single texture can hold that. The build must be tiled exactly as NASA's own products are: CTX everywhere, HiRISE only where the viewer goes close. All numbers in this note are arithmetic from the resolutions above.

Vertical exaggeration - decide once, declare it everywhere

- **True scale (1x):** 800 m of total relief across a 45 km crater is under 2 percent. In a true-scale model the rim reads as a low swell, which is honest but can look "flat" to a judge standing inside it.
- **Diorama / 3D print:** 3x to 5x vertical exaggeration makes the delta cliffs (610 m rim wall [R16]) and the 400 m wide Neretva channel legible. State the factor on the model itself.
- **Simulation / game:** Keep 1x heights but exaggerate slope shading, not geometry, so rover speeds and the 23-degree slope limit [R15] stay truthful.

Elevation graph source: NASA JPL, PIA26476, Jezero Crater Elevation Graph - https://www.jpl.nasa.gov/images/pia26476-jezero-crater-elevation-graph/ [R14]. Open it beside this page: its dotted lake-level lines are the waterlines to engrave on the Margin Unit.

2. Scientific area

Jezero was not chosen for its looks. NASA's mission page records that the landing-site team examined more than 60 candidate locations over a five-year study before Jezero rose to the top, because it combines an ancient lake, a well-preserved river delta and diverse mineralogy in one crater [R1]. The design must preserve that combination: crater floor, western delta, inlet valley, and shoreline unit inside one rim.

2.1 How the lake system worked

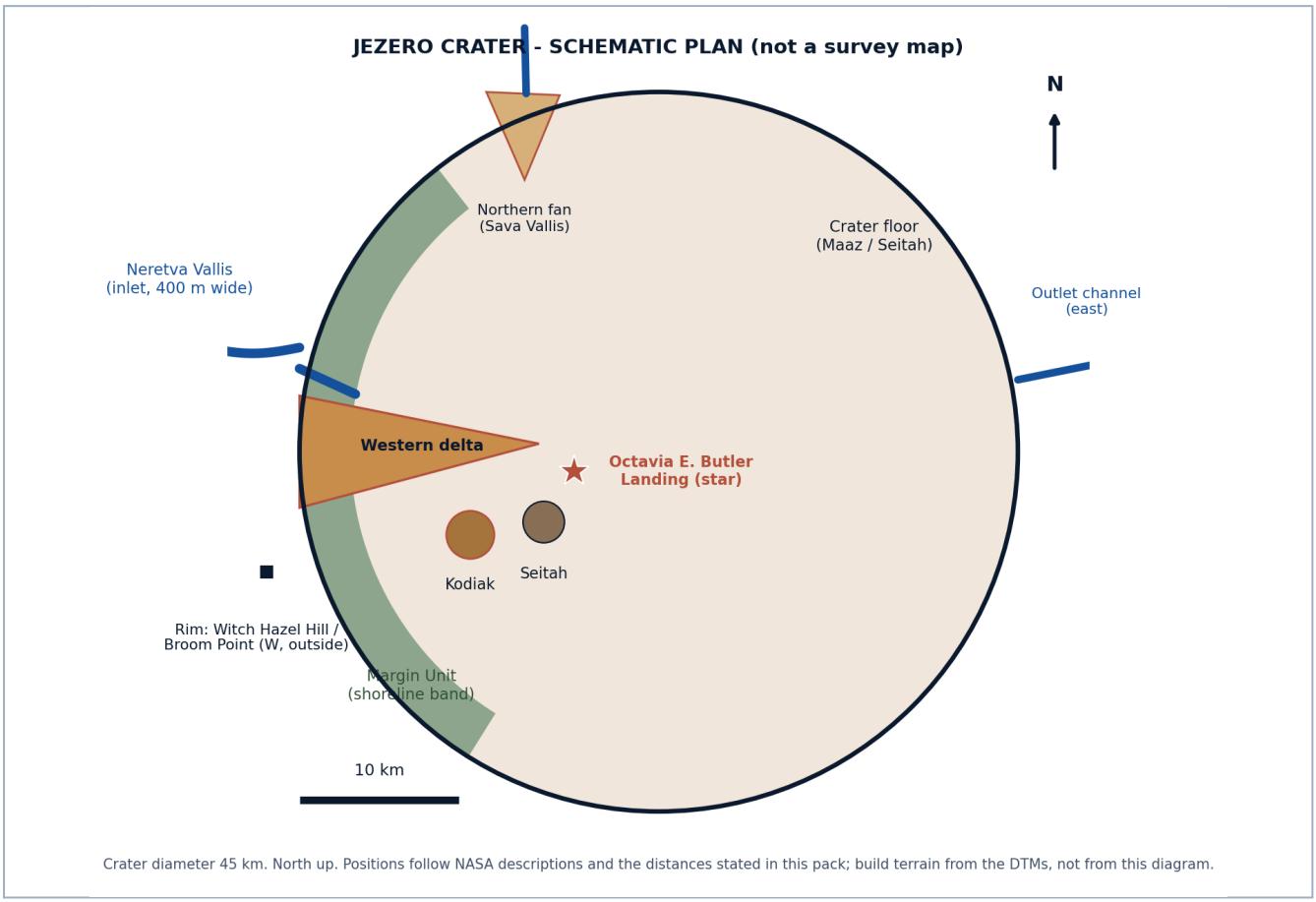
- **Filling:** More than 3.5 billion years ago, river channels spilled over the crater wall and created a lake [R1]. "Jezero" means "lake" in several Slavic languages [R2].
- **Delta building:** A river flowed into a body of water and deposited sediments in a fan shape known as a delta. NASA's landing press kit describes that body of water as about the size of Lake Tahoe [R13].
- **Mineral delivery:** Water carried clay minerals from the surrounding area into the crater lake.
- **Drying and re-wetting:** Jezero's history is the on-again, off-again wet past of Mars; the Margin Unit alone records three separate interactions with water (Section 3.3) [R1][R11].

2.2 The six zones to model - summary (build sheets in Section 2.4)

Zone	What NASA says is there	Design instruction
Crater floor	Perseverance landed here on Feb. 18, 2021 [R1]. NASA reports the rover found two different types of volcanic rock that had interacted with water making up the crater floor [R1].	Flat volcanic plain, locally altered by water. The landing and starting area - keep it open and distinct from the delta sediments.
Octavia E. Butler Landing (Zone 6)	Landing site, Feb. 18, 2021, at 18.4446 N, 77.4509 E [R26], about 1.9 km east of the delta remnants [R31].	The hub and spawn point: a flat Maaz paver clearing, starred on every map in this pack.
Western delta and scarps	Fan-shaped sediment deposit where the river entered the lake. Scarps A-D are its steep front; Kodiak is a detached remnant of the same delta, about 2.2 km from the rover position in PIA24814 [R5][R6].	Layered sedimentary fan with a steep front. Include the boulder-bearing upper scarp layers - boulders up to 1.5 m [R6].
Margin Unit (shoreline)	The geologic area along the inner edge of the rim, on the ancient lake's shoreline. Expected to be sedimentary; Perseverance instead found igneous rock with a complex water record. Explored across approximately 870 feet (265 metres) of elevation; more than 185 bedrock targets analysed with SuperCam [R11].	A sloping shoreline band that changes with elevation: coarse olivine high up; fractured, silica- and carbonate-bearing rock low down (Section 3.3).
Neretva Vallis and Bright Angel	An ancient river valley a quarter-mile (400 metres) wide, carved by water rushing into Jezero. The Bright Angel formation is a set of rocky outcrops on its northern and southern edges, where Perseverance found Cheyava Falls in July 2024 [R12].	Cut a 400 m wide inlet channel through the western rim. Place the Bright Angel clay-and-silt outcrops along its edges - the Cheyava Falls location.
Crater rim	The crater wall the rivers spilled over [R1]. The eastern Margin Unit, near the rim, holds the later mineral veins in Section 3.3 [R11].	A continuous enclosing rim, breached at the inlet valley. At 45 km across [R1], the rim reads as distant terrain from the floor.

2.3 The work area as a map - six zones

The team is not modelling all of Mars. The work area is Jezero's western half - where the river entered, the delta stands, and the rover has actually driven - read against the full 45 km crater for context. The diagram below fixes the layout; the master table fixes the data. Detailed build sheets for each zone follow on the next three pages.



Schematic plan drawn for this pack from NASA's stated diameter, positions and distances. It is a layout diagram, not survey data: terrain geometry must come from the CTX and HiRISE DTMs in Section 1.5.

Zone master table - sample data for all six zones

Zone	Anchor / position	Elevation	Defining size	Palette (design)
Z1 Crater floor	Crater interior; Seitah about 1 km SW of landing [R26]	About -2,600 m; traverse varies ~13 m [R14]	Fills the 45 km bowl [R1]	Pale fractured tan
Z2 Western delta + Kodiak	West wall, at the Neretva mouth; scarps ~2.2 km from the PIA24814 rover position [R6]	Fan top above floor; cliffs at its base [R16]	Boulders to 1.5 m in scarp tops [R6]; 1 km crater in fan [R16]	Layered ochre bands
Z3 Margin Unit	Inner western rim, ancient shoreline [R11]	265 m explored span [R11]	185+ SuperCam bedrock targets [R11]	Green-grey olivine to pale carbonate
Z4 Neretva Vallis + Bright Angel	Channel through the western rim; outcrops on its N and S edges [R12]	Channel floor below rim crest; Bright Angel at the northern wall base [R18]	Channel 400 m wide [R12]; N. Bright Angel exposure at least 1.2 km long [R30]	Light-toned cream outcrops
Z5 Crater rim + Broom Point	Western rim and just outside it [R23]	+300 m to Aurora Park [R15]; +800 m total relief at Lookout Hill [R22]	Rim wall 610 m [R16]; Broom Point stack 75 m thick [R23]	Dark ancient bedrock, bright Broom Point band
Z6 Octavia E. Butler Landing	18.4446 N, 77.4509 E [R26]; ~1.9 km east of delta remnants [R31]	-2,570 m (reported)	Landing 18 Feb 2021 [R1]; start of every traverse	Maaz paver field, rust dust

Palette column is this pack's design assignment for telling zones apart at a glance. It is a design rule, not a NASA measurement - true surface colour comes from Perseverance imagery (Sections 1.5 and 3.4).

2.4 Zone build sheets

Z1. Crater floor - Seitah and Maaz formations

Position	The crater interior. Perseverance landed on the Maaz formation and entered Seitah about 1 km southwest of the landing site [R26].
Elevation	About -2,600 m general floor [R14]; reported traverse values -2,570 m at the landing site to -2,583 m at Rochette, South Seitah. Model relief: almost flat, with low domes - Seitah rocks stand up to 17 m above adjacent Maaz flows in the fold core (mission-team mapping).
Geology	Two volcanic formations. Seitah: olivine-rich cumulate, coarse pyroxene enclosing millimetre olivine grains, aqueously altered. Maaz: basaltic lava flows of interlocking pyroxene and plagioclase (feldspar), with olivine altered to iron-rich serpentine.
Chemistry	Olivine, pyroxene, plagioclase; iron-rich serpentine alteration in Maaz. Carbonate appears where floor rocks met water - keep it patchy, never a floor-wide layer.
Build instruction	A broad, open, traversable plain: fractured light-toned pavers (Maaz) against darker, rougher Seitah outcrops. This is the player's / viewer's starting ground - keep sightlines to the delta and rim open.

Build from: Figure PIA23239 (Section 1.1) for the floor texture; PIA24379 (Section 1.4a) for its flatness against the rim.

Z2. Western delta and Kodiak - Shenandoah, Tenby and Otis Peak

Position	Against the western wall at the mouth of Neretva Vallis. In PIA24814 the scarps and Kodiak sit about 2.2 km from the rover position shown [R5][R6].
Elevation	Fan surface above the floor, ending in the cliffs that define the delta base [R16]. The delta-top map PIA25673 carries a 0-1,000 m scale bar - the fan top is a few kilometres across [R17].
Geology	Layered river-delta sediments. Fan front: Shenandoah formation. Upper fan: Tenby and Otis Peak formations, including a field of metre-scale boulders on the fan top. Scarps A-D are the steep delta front; Kodiak is a detached, flat-topped remnant of the same delta [R5][R6].
Chemistry	Clays and carbonates in the delta sediments [R3]; olivine grains in the fan's clastic rocks came from more than one source rock.
Build instruction	Sculpt visible horizontal layering. Boulder-bearing upper scarp layers (boulders to 1.5 m [R6]), a steep front, and a separate Kodiak butte - never merge Kodiak into the main fan.

Build from: Figures PIA24814 (Section 1.2) and PIA25673 (Section 1.4b).

Z3. Margin Unit - the ancient shoreline

Position	A band along the inner edge of the western rim, between the delta and the rim wall - the shoreline of the ancient lake [R11].
Elevation	Explored across about 265 m of elevation, with more than 185 bedrock targets analysed by SuperCam [R11]. Chemistry changes with height - this gradient is the zone's main design feature.
Geology	Expected from orbit to be sedimentary; found on the ground to be igneous rock altered by water three separate times (Section 3.3). High ground: coarse, crystalline olivine barely touched by water. Low ground: fractured olivine with silica between grains and carbonate-filled fractures standing as ridges [R11].
Chemistry	Olivine (high) -> carbonate + silica (low, below the old waterline). Eastern Margin Unit only: calcium sulfate + fluorite veins about 25 cm thick [R11].
Build instruction	A long sloping bench, not a beach. Grade its material continuously with elevation and engrave the PIA26476 lake-level lines across it [R14].

Build from: Figure PIA24379 (Section 1.4a) for the bench shape; chemistry placement from Section 3.



PIA26336. Credit: NASA/JPL-Caltech. Perseverance in the Neretva Vallis channel on sol 1172 (June 6, 2024), looking toward Bright Angel - the light-toned area at the base of the northern wall. Navigation camera view [R18]. NASA reference: http://science.nasa.gov/photojournal/perseverance-views-bright-angel/

Z4. Neretva Vallis and Bright Angel - the river inlet

Position	The ancient river channel cut through the western rim - the primary inlet that fed the lake. Bright Angel outcrops sit on its northern and southern edges; Cheyava Falls is at the northern edge [R12].
Elevation	Channel floor between rim crest and crater floor. Bright Angel lies at the base of the channel's northern wall [R18].
Geology	Channel 400 m (a quarter-mile) wide [R12]. Bright Angel: light-toned, banded, fine-grained mudstone-to-siltstone outcrops, distinct from the darker Margin Unit beside them; the northern exposure runs at least 1.2 km toward the crater floor [R30].
Chemistry	Clay-and-silt sedimentary rock rich in organic carbon, with sulfur, oxidised iron (rust) and phosphorus [R12]. At Cheyava Falls: white calcium sulfate veins, hematite-red bands, olivine crystals, and the leopard-spot minerals vivianite and greigite [R19].
Build instruction	Cut the channel clean through the rim - it is the only road into the crater. Make Bright Angel visibly paler than everything around it, and place the Cheyava Falls rock (1 m by 0.6 m [R19]) as a findable hero object on the northern side.

Build from: Figure PIA26336 above; spectra in Section 3.4.

Z5. Crater rim and Broom Point - the oldest ground

Position	The western rim crest and the ground just outside it. Rim stops in driving order: Dox Castle, Pico Turquino, Aurora Park, Lookout Hill, Witch Hazel Hill, then Broom Point on the outer rim [R15][R22][R23].
Elevation	+300 m from the floor to the Aurora Park summit, on slopes up to 23 degrees [R15]; Lookout Hill stands 800 m above the traverse's lowest point [R22]; the western rim wall is 610 m high above the delta base [R16].
Geology	Ancient crust predating the crater. The science team divided the rim into 36 map quadrants from HiRISE imagery [R28]. Dox Castle marks the Margin-to-rim transition; Pico Turquino holds ancient fractures that may record hydrothermal activity; Witch Hazel Hill is light-toned layered bedrock [R15]. Broom Point: a 75 m thick stack of six rock types - breccias of angular fragments alternating with pulverised rock dust, glassy impact beads, layers tilted over 80 degrees - more than 3.9 billion years old, built by repeated asteroid impacts and sampled as the Bell Island and Main River cores [R23].
Chemistry	Impact breccias and melt-bearing fragments; possible hydrothermal fracture minerals at Pico Turquino are a target to test, not a confirmed find [R15].
Build instruction	The rim must read as older and rougher than everything inside the crater: broken, tilted strata at Broom Point, a bright band for its layers, and a crest viewpoint that reveals Nili Planum outside [R22].

Build from: Figure PIA24379 (Section 1.4a) for the wall; Section 1.6 for the climb profile.

Z6. Octavia E. Butler Landing - the operations hub

Position	18.4446 degrees N, 77.4509 degrees E [R26], on the Maaz formation at the western edge of the crater-floor unit. About 1.9 km east of the remnants of the river delta [R31], and about 2.2 km southeast of the main delta deposits.
Elevation	-2,570 m (reported from the first traverse) - 30 m above the general floor value, on nearly flat ground chosen for a safe landing.
Geology	Maaz paver field: flat, polygonal-fractured, light-toned paver rocks with darker rocks standing proud of the surface near the site. Every traverse in this pack starts here: floor campaign south to Seitah, then west to the delta, the Margin Unit, Neretva Vallis and the rim [R29].
Chemistry	Basaltic Maaz mineralogy (Zone 1). No special chemistry - its value is positional, not mineralogical.
Build instruction	Model it as the hub and spawn point: a small, flat, recognisable clearing with the traverse lines radiating out exactly as drawn in PIA24379 and the live location map [R16][R29]. Mark it with a star, as the schematic plan does.

Build from: Figures PIA24379 (Section 1.4a) and PIA25673 (Section 1.4b, white traverse).

3. Chemicals and minerals

Every entry below was detected either from orbit by an MRO or Mars Odyssey instrument, or on the ground by Perseverance. The "where" column is the placement rule for the design: a mineral placed in the wrong zone is a factual error, not an artistic choice.

Mineral / chemical	What NASA reports	Where in Jezero	Detected by / source
Clay minerals (phyllosilicates, including smectite-type clays)	Water carried clay minerals into the crater lake, and the delta sediments contain clays. On the ground, the Bright Angel sedimentary rocks are composed of clay and silt, which on Earth are excellent preservers of past microbial life [R1][R3][R12].	Delta sediments; Bright Angel formation on the edges of Neretva Vallis.	CRISM from orbit [R3]; Perseverance at Bright Angel [R12].
Carbonates (Mg-Fe carbonate)	Delta sediments contain carbonates [R3]. Jezero sits inside one of the largest exposures of carbonate on Mars, and orbiters detected strong carbonate signals in the Margin Unit. There, carbonate formed when carbon-dioxide-rich groundwater reacted with olivine, leaving carbonate-filled fractures that now stand as ridges at low elevations [R11].	Delta sediments; Margin Unit, strongest at low elevations.	CRISM from orbit [R3]; SuperCam on Perseverance [R11].
Olivine	Made of magnesium and iron. In the Margin Unit it is coarse-grained and crystalline - it formed in a body of magma deep underground, cooled slowly enough for large grains to grow, and reached the surface only after the rock above it eroded away [R11].	Margin Unit at higher elevations, with almost no sign water touched it [R11].	Orbital detection, confirmed by SuperCam [R11].

3. Chemicals and minerals (continued) - the Margin Unit alteration minerals

Mineral / chemical	What NASA reports	Where in Jezero	Detected by / source
Silica	Lower in the Margin Unit, olivine grains are fractured with silica between them. Turning olivine into carbonate can leave silica behind, and NASA reports more silica in rocks that sat below the lake's water line. With carbonate, it is a signpost in the search for ancient life, because the olivine-water reaction can release hydrogen, a food source for some microbes [R11].	Margin Unit at low elevations, below the ancient water line.	SuperCam on Perseverance [R11].
Calcium sulfate	Found with fluorite in mineral veins about 10 inches (25 centimetres) thick at one location in the eastern Margin Unit, created by the last of the three water events there [R11].	Eastern Margin Unit only - vein material, not a surface-wide layer.	SuperCam on Perseverance [R11].
Fluorite	Found in the same approximately 25 cm thick eastern Margin Unit veins. NASA notes fluorite typically forms when hot water circulates through volcanic rocks, so its presence reveals a later, heated underground-water event [R11].	Eastern Margin Unit veins only.	SuperCam on Perseverance [R11].

3.1 Chemicals and minerals (continued) - the Bright Angel assemblage

Mineral / chemical	What NASA reports	Where in Jezero	Detected by / source
Organic carbon	The Bright Angel sedimentary rocks are rich in organic carbon, together with sulfur, oxidised iron (rust) and phosphorus. In Cheyava Falls, organic carbon occurs with the leopard-spot minerals in Section 3.2 [R12].	Bright Angel formation, Neretva Vallis edges.	PIXL and SHERLOC on Perseverance [R12].
Vivianite (hydrated iron phosphate)	One of two iron-rich minerals carrying the signature of the leopard spots in Cheyava Falls. On Earth, vivianite is frequently found in sediments, peat bogs and around decaying organic matter [R12].	Cheyava Falls rock, Bright Angel formation.	PIXL and SHERLOC [R12].
Greigite (iron sulfide)	The second leopard-spot mineral. NASA notes that certain forms of microbial life on Earth can produce greigite - and that both vivianite and greigite can also be generated without life [R12].	Cheyava Falls rock, Bright Angel formation.	PIXL and SHERLOC [R12].

Formulas: NASA's pages consulted here name these minerals and, for olivine, state its magnesium-and-iron composition; they do not print formulas for vivianite or greigite, and none has been invented to fill that gap. If the design needs formulas on screen, take them from the PDS instrument data in Section 1.5, not from a general website.

3.2 Cheyava Falls and Sapphire Canyon - the potential biosignature

- **Rock:** Cheyava Falls, an arrowhead-shaped rock measuring 3.2 feet by 2 feet (1 metre by 0.6 metres), found in July 2024 in the Bright Angel formation [R12].
- **Sample:** "Sapphire Canyon," the core Perseverance collected from Cheyava Falls - one of 27 rock cores the rover had collected since landing in February 2021 at the time of NASA's announcement [R12].
- **Features:** Coloured spots and, at higher resolution, a distinct pattern of minerals arranged into reaction fronts the team called leopard spots, carrying the signature of vivianite and greigite [R12].
- **Energy chemistry:** The combination of chemical compounds in the Bright Angel formation could have been a rich source of energy for microbial metabolisms, and the two leopard-spot minerals appear to have formed by electron-transfer reactions between the sediment and organic matter [R12].
- **Status:** After a yearlong peer review, the journal *Nature* published the validated results in September 2025: the Sapphire Canyon sample contains potential biosignatures [R1][R12].

> **Do not overstate this in the design.** NASA defines a potential biosignature as a substance or structure that might have a biological origin but requires more data or further study before a conclusion can be reached about the absence or presence of life [R12]. NASA's project scientist states that while abiotic explanations are less likely given the findings, they cannot be ruled out. Any label, caption or narration in the build must say "potential biosignature," never "life found on Mars."

3.3 The Margin Unit's three water events - build them in this order

NASA's Margin Unit study, based on Perseverance data and published in the journal *Communications Earth & Environment*, found that the rocks interacted with water on at least three separate occasions. The team can determine the sequence, but not the age, of each event [R11]. If the design shows alteration, weathering or a timeline, it must follow this order:

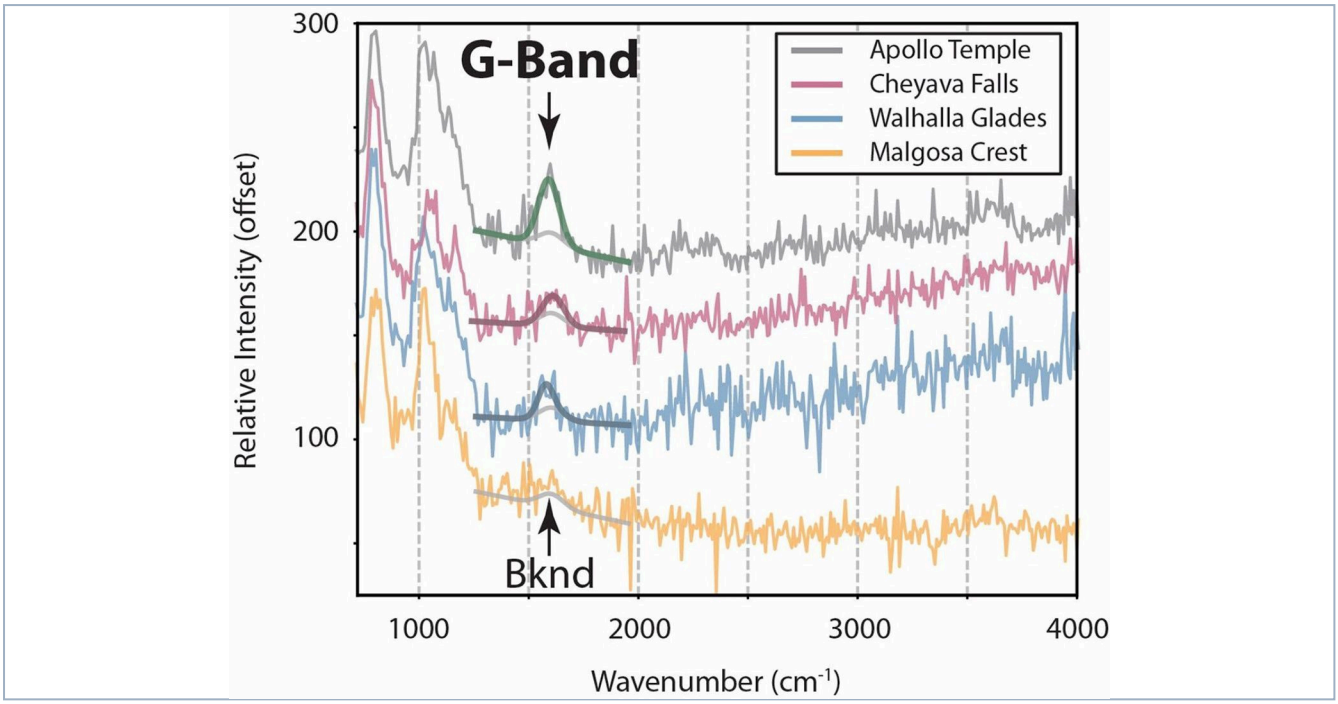
Order	What happened	What it left behind
1st	Carbon-dioxide-rich groundwater reacted with olivine.	Ridges of carbonate running through fractures in bedrock at low elevations. The carbonate-filled fractures are left standing today as the softer rock around them wears away [R11].
2nd	A water event that may have been related to the lake that once existed in the crater. Rocks that sat below the water line were altered most.	More silica in the rocks below the water line, from olivine turning into carbonate [R11].
3rd	Heated underground water circulated through the volcanic rocks.	Mineral veins about 25 cm thick in the eastern Margin Unit, containing calcium sulfate and fluorite [R11].

Reading the evidence correctly - three cautions for the developer

- **Diameter - NASA gives two values.** The Mars 2020 mission page and the launch press kit both state 28 miles (45 kilometres) [R1][R2]. The caption of the THEMIS image PIA26140 states 47 km (29 miles) [R4]. Use 45 km as the design scale (two NASA mission sources against one image caption), and record the 47 km caption value in the project credits so a judge who checks PIA26140 finds it acknowledged, not hidden.
- **False colour - materials, not appearance.** PIA26140 and the CRISM overlay in PIA23239 encode mineral and surface differences. Their greens, blues and purples must be translated into material zones, not copied as the ground's literal colour (Section 1.3) [R3][R4]. Ground truth for appearance is the Perseverance raw imagery (Section 1.5).
- **Expected versus found at the Margin Unit.** From orbit, the Margin Unit looked sedimentary and carbonate-rich. On the ground it is igneous rock altered by water three times [R11]. Model what Perseverance found, not what the orbital expectation was - that surprise is itself one of the site's main science stories.

3.4 Sample records - what a real Jezero data point looks like

The tables in Sections 1 to 3 are compiled data. This page shows the primary records behind the headline claims, so the developer can style in-world data displays, sample labels and museum captions on real NASA records instead of invented ones.



PIA26639. Credit: NASA/JPL-Caltech. SHERLOC Raman spectra from Bright Angel targets. The G-band signal indicates organic molecules, which geological as well as biological processes can create; the Malgosa Crest trace shows background only [R21].

Record	Sample data - exactly as NASA reports it
PIA26639 - SHERLOC at Bright Angel	Instrument: SHERLOC, on the end of Perseverance's robotic arm. Targets: Apollo Temple, Cheyava Falls, Walhalla Glades, Malgosa Crest. Cheyava Falls (second trace) was later determined to contain a potential biosignature. Published 2025-09-10 [R21]. Reference: https://science.nasa.gov/photojournal/perseverances-sherloc-finds-organic-molecules-in-bright-angel/
PIA26368 - Cheyava Falls close-up	Instrument: WATSON camera, part of SHERLOC. Captured July 18, 2024, sol 1212. Rock 1 m by 0.6 m; white calcium sulfate veins run its length; reddish hematite bands; millimetre leopard spots with dark rims; green olivine crystals in the white material at its sides. Cored July 21, 2024 as the rover's 22nd rock sample - Sapphire Canyon [R19]. Reference: https://science.nasa.gov/resource/perseverance-finds-a-rock-with-leopard-spots/
PIA26369 - Bright Angel panorama	Instrument: Mastcam-Z. Captured June 12, 2024, sol 1178; 346 individual images stitched into a 360-degree panorama. Cheyava Falls sits slightly right of centre, about 110 m from the rover's position [R20]. Reference: https://science.nasa.gov/resource/perseverance-captures-bright-angel-in-360-degrees/

PIA26368 and PIA26369 are catalogued here with full metadata and links rather than embedded prints, because NASA's pages for them offer the originals at full resolution (the panorama's main version is 53,970 by 13,199 pixels [R20]) - download those two from their NASA pages; every other image in this pack is embedded.

4. References - NASA first, supplements labelled

Every link below was opened and checked against the fact it supports. Cite the PIA number and the NASA page in the Space Apps project credits exactly as listed here.

Ref	NASA source	Used for
[R1]	NASA Science - Mars 2020: Perseverance Rover (mission page) https://science.nasa.gov/mission/mars-2020-perseverance/	Crater size and location, ancient lake, landing-site selection, mission objectives, Sapphire Canyon status.
[R2]	NASA JPL - Mars 2020 Perseverance Launch Press Kit: Landing Site https://www.jpl.nasa.gov/news/press_kits/mars_2020/launch/mission/landing_site/	About 18 degrees latitude, 77 degrees longitude; 45 km crater; lake 3.5 billion years ago; carbonate-rich areas; five kinds of rock.
[R3]	NASA Science Photojournal - PIA23239, Jezero Crater, Mars 2020's Landing Site (2019-05-15) https://science.nasa.gov/photojournal/jezero-crater-mars-2020s-landing-site/ Catalogue: https://photojournal.jpl.nasa.gov/catalog/PIA23239	Base-map image, CRISM + CTX instruments and credits, clays and carbonates in delta sediments.
[R4]	NASA Science Photojournal - PIA26140, Jezero Crater - False Color (2023-09-29) https://science.nasa.gov/photojournal/jezero-crater-false-color/	THEMIS image and full metadata: orbit 95046, latitude 18.449, longitude 77.436, VIS, captured 2023-05-19 09:28; 47 km caption value; false-colour caution.
[R5]	NASA Science Photojournal - PIA24814, Jezero Crater's Kodiak and Scarps (2021-10-07) https://science.nasa.gov/photojournal/jezero-craters-kodiak-and-scarps/	HiRISE delta image, Kodiak, scarps A-D, rover position, credits.
[R6]	NASA JPL - NASA's Perseverance Sheds More Light on Jezero Crater's Watery Past https://www.jpl.nasa.gov/news/nasas-perseverance-sheds-more-light-on-jezero-craters-watery-past/	Scarp and Kodiak distances (about 2.2 km), boulders up to 1.5 m, delta stratigraphy confirmation.
[R7]	NASA Planetary Data System - Imaging Node Data Portal https://pds-imaging.jpl.nasa.gov/portal/	MRO archive access: HiRISE, CTX and MARCI product search and data volumes.

4. References (continued)

Ref	NASA source	Used for
[R8]	NASA PDS Geosciences Node - Mars 2020 Mission data page https://pds-geosciences.wustl.edu/missions/mars2020/	Perseverance instrument bundles, PDS release schedule, Sample Dossier.
[R9]	USGS Astrogeology - Mars 2020 TRN Context Camera Orthorectified Image Mosaic (NASA MRO data, NASA Mars 2020 product) https://astrogeology.usgs.gov/search/map/Mars/Mars2020/JEZ_ctx_B_soc_008_orthoMosaic_6m_Eqc_latTs0_lon0	6 m/pixel CTX mosaic and DTM, coverage bounds.
[R10]	USGS Astrogeology - Mars Sample Return TRN HiRISE DTM and Orthoimage Mosaics (NASA JPL / USGS, DOI 10.5066/P13CPYYU) https://astrogeology.usgs.gov/search/map/mars-sample-return-terrain-relative-navigation-hirise-dtm-mosaic	HiRISE elevation and orthoimage mosaics, alignment accuracies, download sizes.
[R11]	NASA - NASA Discovery Reveals Complex Water Systems on Early Mars (2026-09-21) https://www.nasa.gov/solar-system/planets/mars/nasa-discovery-reveals-complex-water-systems-on-early-mars/	Margin Unit: olivine, carbonate, silica, calcium sulfate, fluorite; 265 m of elevation; 185+ SuperCam targets; three water events.
[R12]	NASA News Release - NASA Says Mars Rover Discovered Potential Biosignature Last Year (2025-09-09) https://www.nasa.gov/news-release/nasa-says-mars-rover-discovered-potential-biosignature-last-year/	Cheyava Falls, Sapphire Canyon, Bright Angel composition, vivianite and greigite, Neretva Vallis width, potential-biosignature definition.
[R13]	NASA JPL - Mars 2020 Perseverance Landing Press Kit: Introduction https://www.jpl.nasa.gov/news/press_kits/mars_2020/landing/	Lake about the size of Lake Tahoe; delta and lake deposits as collectors of organic molecules and potential signs of microbial life.

4. References (continued) - added in the expanded edition

Ref	Source	Used for
[R14]	NASA JPL - PIA26476, Jezero Crater Elevation Graph https://www.jpl.nasa.gov/images/pia26476-jezero-crater-elevation-graph/	Crater about 2,600 m below reference level; SuperCam sampling sites vs elevation; ancient lake levels; rim ascent jump.
[R15]	NASA - NASA's Perseverance Rover to Begin Long Climb Up Martian Crater Rim (2024-08-14) https://www.nasa.gov/missions/mars-2020-perseverance/perseverance-rover/nasas-perseverance-rover-to-begin-long-climb-up-martian-crater-rim/	+300 m climb to Aurora Park; slopes to 23 degrees; Pico Turquino and Witch Hazel Hill targets.
[R16]	NASA Science Photojournal - PIA24379, Possible Path for Perseverance Rover (2021-01-27) http://science.nasa.gov/photojournal/possible-path-for-perseverance-rover/	Route perspective image; rim 610 m high; CTX mosaic 6 m/pixel; 1 km reference crater.
[R17]	NASA Science Photojournal - PIA25673, Map of Perseverance's Delta Top Campaign (2022-12-15) https://science.nasa.gov/photojournal/map-of-perseverances-delta-top-campaign/	Delta-top working map; route and waypoint conventions; 1 km scale bar.
[R18]	NASA Science Photojournal - PIA26336, Perseverance Views 'Bright Angel' (2024-06-13) http://science.nasa.gov/photojournal/perseverance-views-bright-angel/	Navcam view in Neretva Vallis, sol 1172; Bright Angel at the northern wall base.
[R19]	NASA Science - PIA26368, Perseverance Finds a Rock with 'Leopard Spots' (2024-07-25) https://science.nasa.gov/resource/perseverance-finds-a-rock-with-leopard-spots/	Cheyava Falls WATSON record: sol 1212, size, veins, hematite, olivine, 22nd sample.
[R20]	NASA Science - PIA26369, Perseverance Captures 'Bright Angel' in 360 Degrees (2024-07-25) https://science.nasa.gov/resource/perseverance-captures-bright-angel-in-360-degrees/	Mastcam-Z panorama: sol 1178, 346 images, Cheyava Falls 110 m away, full-res size.
[R21]	NASA Science Photojournal - PIA26639, Perseverance's SHERLOC Finds Organic Molecules in 'Bright Angel' (2025-09-10) https://science.nasa.gov/photojournal/perseverances-sherloc-finds-organic-molecules-in-bright-angel/	SHERLOC Raman spectra and G-band organic signal for four Bright Angel targets.
[R22]	NASA Science - Perseverance Blasts Past the Top of Jezero Crater Rim (2024-12-19) https://science.nasa.gov/blog/perseverance-blasts-past-the-top-of-jezero-crater-rim/	Lookout Hill 800 m above the traverse's lowest point; Dox Castle and Pico Turquino passed; Witch Hazel Hill / South Arm.
[R23]	NASA - NASA's Perseverance Rover Reads Record of Ancient Mars Impacts (2026-07-15) https://www.nasa.gov/solar-system/planets/mars/nasas-perseverance-rover-reads-record-of-ancient-mars-impacts/	Broom Point member: 75 m thick, six rock types, over 3.9 billion years old, Bell Island and Main River samples.

4. References (continued)

Ref	Source	Used for
[R24]	USGS Astrogeology - Mars 2020 Terrain Relative Navigation CTX DTM Mosaic (NASA MRO data) https://astrogeology.usgs.gov/search/map/mars_2020_terrain_relative_navigation_ctx_dtm_mosaic	CTX DTM 20 m/pixel; vertical registration 3.8 m; Mars 2020 IAU sphere radius 3,396,190 m.
[R25]	USGS - NASA Mars Perseverance 2020 Terrain Relative Navigation Mosaics (video notes) http://www.usgs.gov/media/videos/nasa-mars-perseverance-2020-terrain-relative-navigation-mosaics	CTX mosaic 6 m/pixel and HiRISE mosaic 25 cm/pixel; HiRISE mosaic used for landing-ellipse hazards.
[R26]	Bell et al. (2022), Science Advances - Mastcam-Z results from Perseverance in Jezero Crater, via NASA NTRS https://ntrs.nasa.gov/api/citations/20230016428/downloads/Bell%20et%20al%202022sciadv.pdf?attachment=true	Landing coordinates 18.4446 N, 77.4509 E; crater about 45 km, 3.7 to 4.1 billion years old; Seitah about 1 km southwest of the landing site.
[R27]	DLR - The topography of Jezero Crater, landing site of NASA's Mars 2020 mission (HRSC supplement, non-NASA) https://www.dlr.de/en/latest/news/2020/03/20200929_topography-of-jezero-crater	Floor about 2,700 m below the areoid in HRSC topography; lake at least 250 m deep, about 1,340 sq km. Labelled supplement wherever used.
[R28]	NASA Science - Reaching New Heights to Unravel Deep Martian History! (2024-09-16) https://science.nasa.gov/blog/reaching-new-heights-to-unravel-deep-martian-history/	Rim over 300 m above the landing site; Dox Castle first rim stop; rim divided into 36 HiRISE map quadrants.
[R29]	NASA Science - Perseverance Rover Location Map (live) https://science.nasa.gov/mission/mars-2020-perseverance/location-map/	Latest rover position and traverse on the HiRISE colour base map; one dot per drive, labelled by sol.
[R30]	AGU / JGR - Abrasion Patch Dehydration Experiment at Bright Angel, Jezero Crater (2025) https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009243	Bright Angel northern exposure at least 1.2 km long; mudstone-to-siltstone lithologies; Margin Unit contact.
[R31]	Beegle et al. - SHERLOC: Results of the First 350 Sols of Operations, via NASA NTRS https://ntrs.nasa.gov/api/citations/20220002605/downloads/BEEGLE_SHERLOC%20COSPAR%202022.pdf	Octavia E. Butler landing site about 1.9 km east of the delta remnants.

Sources R24, R25, R27 and R30 are the labelled supplements the team agreed to allow: USGS products built from NASA MRO data, DLR/HRSC topography, and a peer-reviewed mission-team journal article. Every other source in this pack is NASA.

5. Map build specification - work only from this pack

This section turns the research into build settings. Anything marked "computed" is arithmetic from values in this pack; anything marked "design rule" is this pack's instruction, not a NASA measurement. If a setting is not in this section or the sections it points to, the developer does not need it.

5.1 Coordinate frame and scale

Projection	Equirectangular, latitude of true scale 0, longitude origin 0 - the "Eqc_latTs0_lon0" frame named in the Mars 2020 product file names [R9].
Elevation datum	DTM heights are in metres referenced to the Mars 2020 IAU sphere, radius 3,396,190 m [R24]. Mission reporting uses the areoid (Mars "sea level") - do not mix the two datums in one model without an offset.
Map centre	Latitude 18.449, longitude 77.436 (THEMIS image centre [R4]). Landing anchor: 18.4446 N, 77.4509 E [R26].
Map extent	Crater diameter 45 km [R1]. CTX mosaic coverage: latitude 18.211 to 18.7212, longitude 77.1605 to 77.6992 [R9].
Texture budget (computed)	45 km at 6 m/pixel (CTX) = 7,500 pixels across - one 8k texture holds the whole crater at CTX fidelity. At 25 cm/pixel (HiRISE) it would take 180,000 pixels across - tile HiRISE areas separately (Section 1.6).
Scale bar	Copy PIA25673: a 0-1,000 m bar and a north arrow on every map view [R17]. On the schematic plan (Section 2.3) the bar is 10 km.

5.2 Layer stack - build in this order

#	Layer	Source in this pack
1	Terrain mesh	CTX DTM 20 m/pixel everywhere; HiRISE DTM 1 m in hero areas (Section 1.6) [R24][R10]
2	Base texture	CTX mosaic 6 m/pixel (Section 1.5) [R9]
3	Detail texture	HiRISE 25 cm/pixel tiles: delta front, landing area, Bright Angel (Section 1.6) [R25]
4	Mineral zoning	CRISM overlay in PIA23239 + the zone chemistry rules (Sections 1.1, 3) [R3]
5	Material variation	THEMIS false colour PIA26140 - zoning only, never literal colour (Section 1.3) [R4]
6	Ground truth check	Perseverance raw images + PIA26336 / PIA26369 views (Sections 1.5, 2.4, 3.4) [R18][R20]
7	Routes and labels	Traverse lines from PIA24379 and PIA25673 conventions; live check on the location map (Section 1.4) [R16][R17][R29]

5.3 Legend, labels and figure index

Legend entry	Design colour	Where it applies
Crater floor (Z1)	Pale fractured tan	Maaz pavers lighter, Seitah outcrops darker and rougher
Western delta (Z2)	Layered ochre bands	Visible strata on scarps A-D and Kodiak; boulders on top layers
Margin Unit (Z3)	Green-grey to pale carbonate	Grade continuously with elevation over the 265 m span [R11]
Neretva + Bright Angel (Z4)	Channel shadow blue-grey; outcrops cream	Bright Angel must be the palest rock in the scene [R18]
Rim + Broom Point (Z5)	Dark ancient bedrock; bright Broom Point band	Tilted strata over 80 degrees at Broom Point [R23]
Landing hub (Z6)	Rust dust, star marker	Flat Maaz clearing at 18.4446 N, 77.4509 E [R26]

Labels that must appear on the finished map

Jezero Crater - Isidis Planitia (west edge) - Neretva Vallis - Sava Vallis - northern fan - western delta - Scarps A, B, C, D - Kodiak - Margin Unit - Bright Angel - Cheyava Falls - crater floor (Maaz / Seitah) - Octavia E. Butler Landing - Dox Castle - Pico Turquino - Aurora Park - Lookout Hill - Witch Hazel Hill - Broom Point - Nili Planum (outside the rim) - eastern outlet channel.

Figure index - every image embedded in this pack

Figure	Subject	Used for
PIA23239	Whole crater, CRISM + CTX	Base map and mineral zoning (Section 1.1)
PIA24814	Kodiak and scarps, HiRISE	Delta-front sculpting (Section 1.2)
PIA26140	Western half, THEMIS false colour	Material zoning (Section 1.3)
PIA24379	Possible-path perspective, CTX	3D shape and rim height (Section 1.4a)
PIA25673	Delta Top Campaign map	Working-map conventions and delta texture (Section 1.4b)
PIA26336	Neretva Vallis toward Bright Angel, Navcam	Ground-level channel view (Section 2.4)
PIA26639	SHERLOC spectra, Bright Angel	Sample-data displays (Section 3.4)
Pack charts	Elevation anchors; schematic plan	Compiled from the cited values only (Sections 1.6, 2.3)

Developer handoff checklist - expanded edition

[ ]	Terrain built from the CTX DTM [R24] refined with HiRISE DTMs [R10] - elevations anchored to Section 1.6, with the vertical-exaggeration factor declared on the model.
[ ]	Six zones present and placed as in Sections 2.3 and 2.4: Z1 crater floor (Seitah / Maaz), Z2 western delta with scarps A-D and Kodiak, Z3 Margin Unit, Z4 Neretva Vallis (400 m wide) with Bright Angel, Z5 rim with Broom Point, Z6 Octavia E. Butler Landing hub at 18.4446 N, 77.4509 E [R26].
[ ]	Map conventions copied from PIA25673: north arrow, 1 km scale bar, white = driven route, black = planned route, dots = waypoints [R17]. All labels from the Section 5.3 list are on the map.
[ ]	Minerals placed by zone and elevation as in Section 3 - calcium sulfate and fluorite only as eastern Margin Unit veins; vivianite and greigite only at Cheyava Falls; Margin Unit chemistry graded over its 265 m span [R11].
[ ]	Scale set to 45 km crater diameter [R1][R2], with the 47 km PIA26140 caption value noted in the credits [R4]. Slopes on the rim climb at or below the 23-degree maximum NASA reports [R15].
[ ]	Colours checked against Perseverance imagery (PIA26336, PIA26369, raw images); false-colour products used for material zoning only [R3][R4][R18][R20].
[ ]	All on-screen wording says "potential biosignature" for Sapphire Canyon, with no claim that life was found [R12]. Sample labels, if any, follow the real records in Section 3.4.